

Experimental Design

RNA-seq, ChIP-seq, ATAC-seq, and sample planning

RNA-Seq

Power Calculation prior to experiment
Paired-end sequencing always recommended
Low tumor purity requires more reads

Study Type	Sequencing Depth	Biological Replicates	Technical Replicates	Why?
Differential Expression (Moderate to large difference)	15M-25M	Ideally 6+, minimum of 3	Ideally 3, minimum 2	15-25M reads captures differences between highly expressed genes. - More replicates = more statistical power. - Technical replicates account for technical variability.
Differential Expression (Small difference)	30M-50M	Ideally 6+	Ideally 3, minimum 2	When expecting a smaller effect size between conditions, more replicates and more depth provides more statistical power.
Transcript Isoform Discovery/Splicing analysis	100M ideally, 50M minimum (Paired-end)	Ideally 6+,	Ideally 3, minimum 2	- Paired end allows for detection of splicing events. - High sequencing depth and many replicates needed since measuring transcript abundance instead of gene expression
Differential Expression (lncRNA/low expressing genes)	100M ideally, 50M minimum	Ideally 6+	Ideally 3, minimum 2	High sequencing depth needed since expression is low in targeted genes

ChIP-Seq

Power Calculation prior to experiment

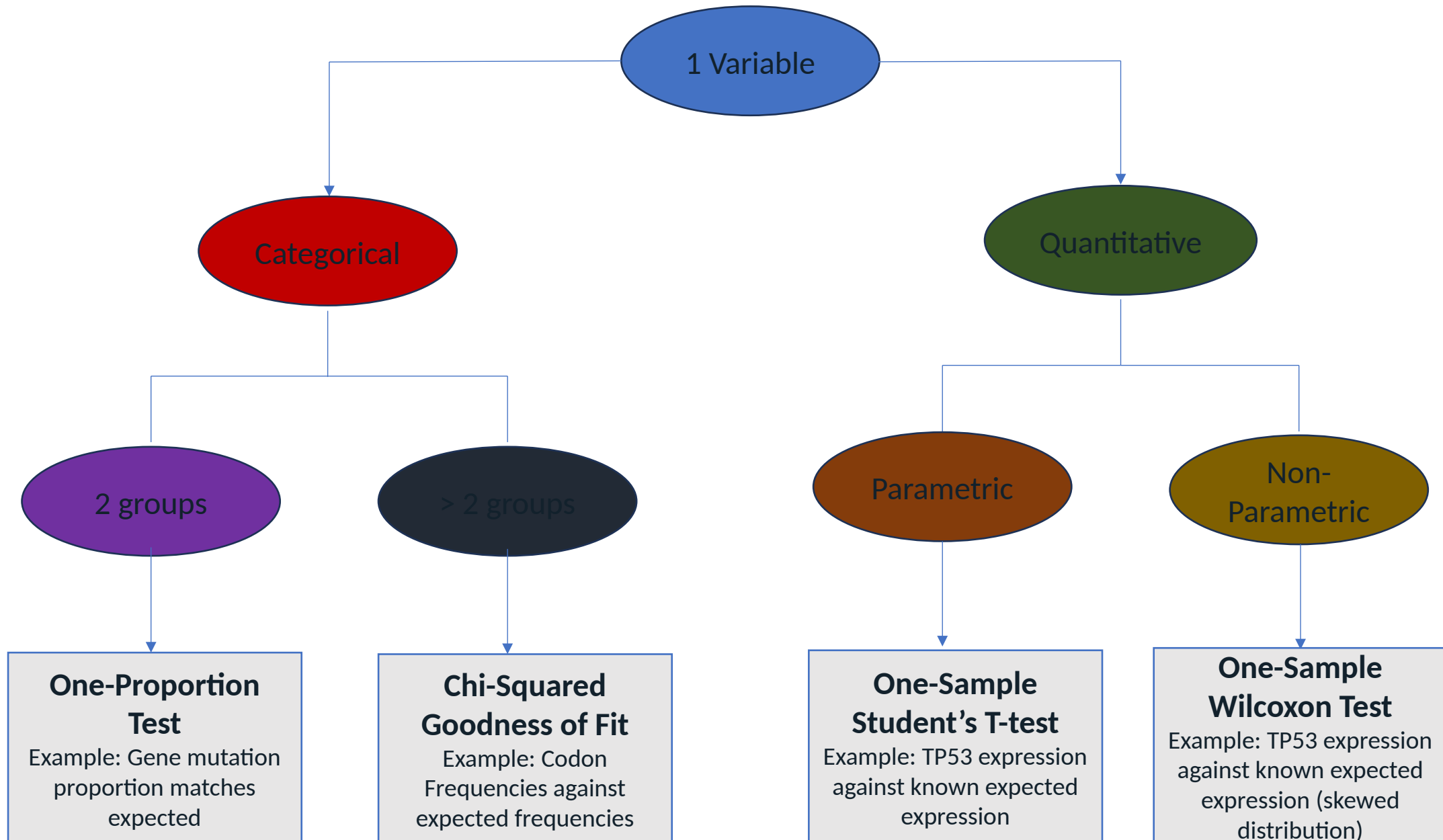
Input depth should be twice as high as ChIP depth

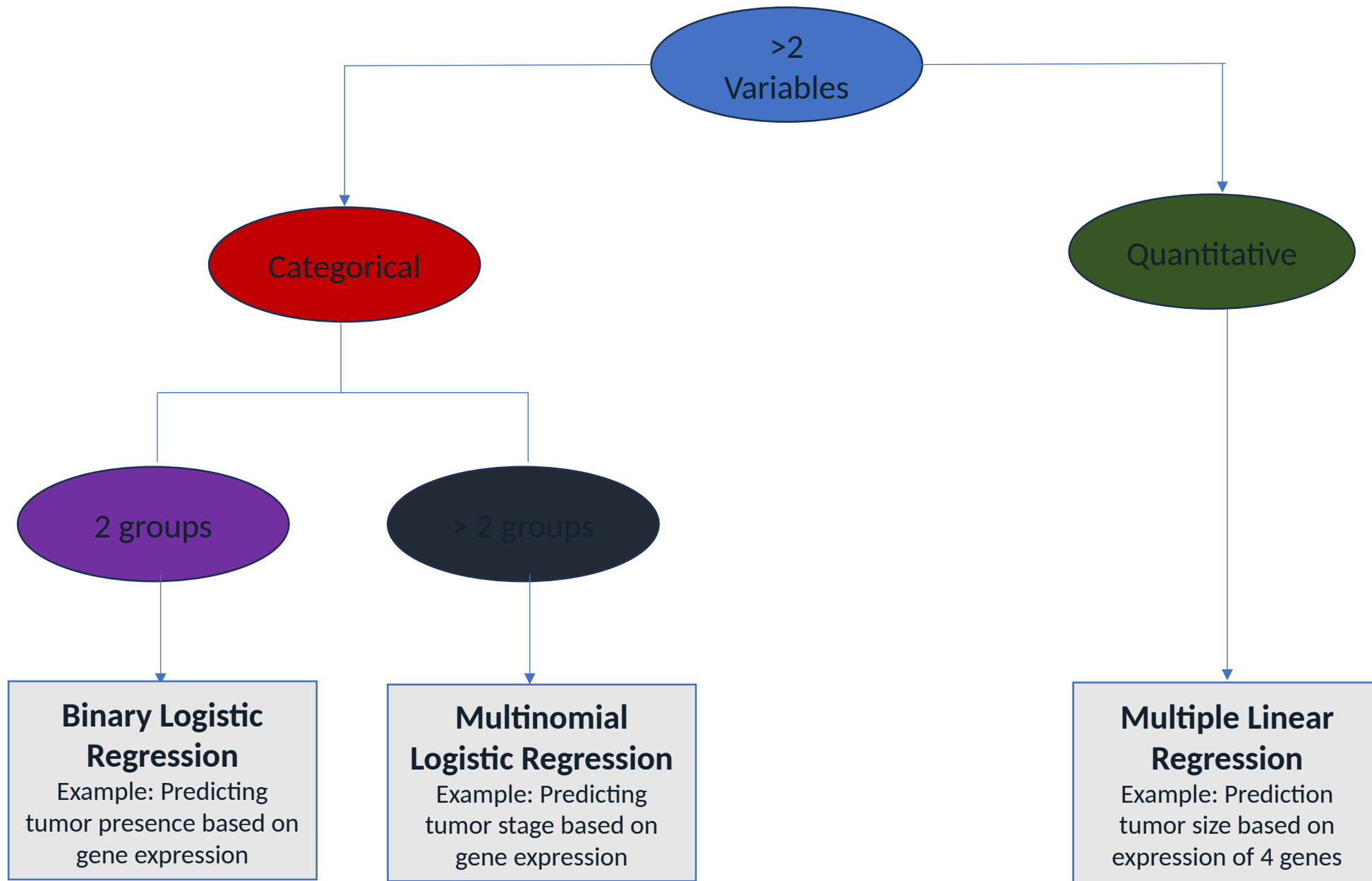
Study Type	Peak Type	Sequencing Depth	Biological Replicates	Technical Replicates	Input Required?	Why?
TF Binding	Narrow	20M-30M	Ideally 3+	Minimum 2	Yes	- TF binding sites typically produce sharp peaks and require lower sequencing depth than other ChIP-seq protocols. - Biological and technical replicates allow for stronger statistical power and control for technical variability
Broad Histone Marks	Broad	50M-100M	Ideally 3+	Minimum 2	Yes	Broad histone marks cover large regions, requiring higher sequencing depth and more samples for statistical power
Narrow Histone Marks	Narrow	20M-50M (depending on histone mark)	Ideally 3+	Minimum 2	Yes	Narrow histone marks cover regions of smaller regions than broad sequencing depth, requiring lower sequencing depth and less samples for statistical power than broad histone marks
Subtle/low binding patterns	Broad or narrow	75M+	Ideally 3+	Minimum 2	Yes	Subtle peaks require a very high sequencing depth since there won't be as many reads mapping to these peaks.

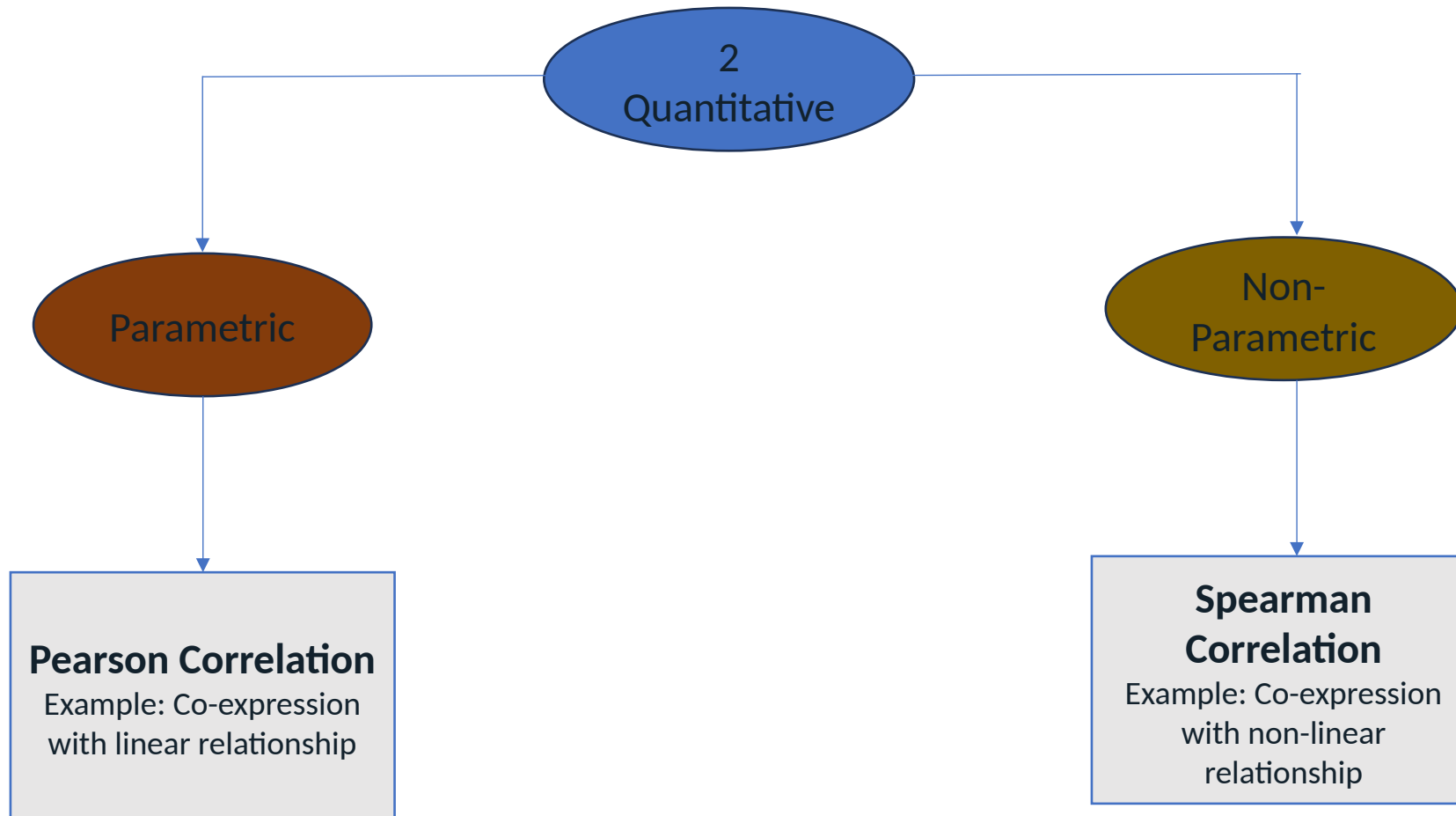
ATAC-Seq

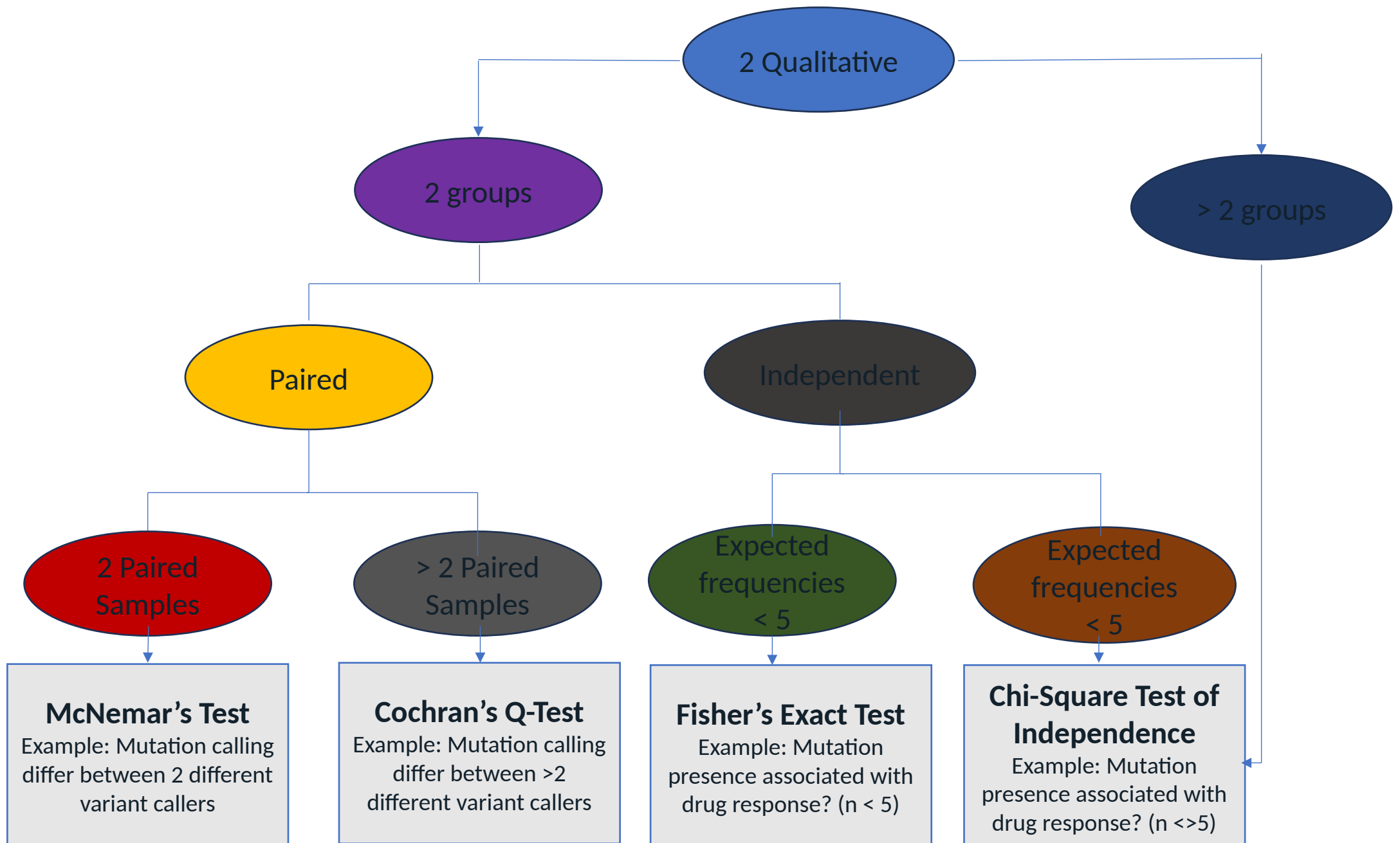
Power Calculation prior to experiment
 Fragment length < 100 = Nucleosome free
 Fragment length ~ 150-200 = Mono-nucleosome bound
 Fragment length >= 300 = Di/Multi-nucleosome bound

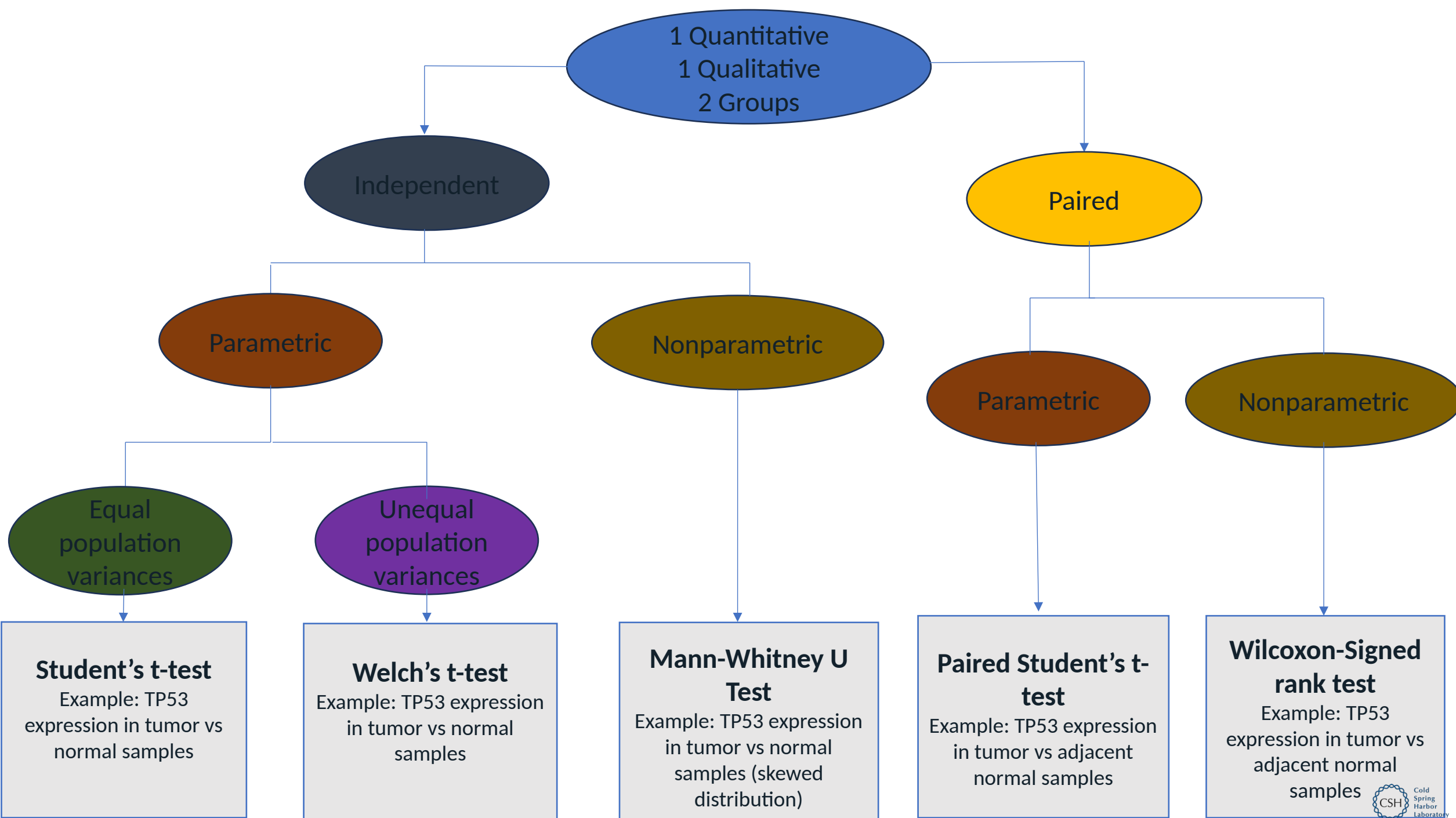
Study Type	Peak Type	Sequencing Depth	Biological Replicates	Technical Replicates	Input Required?	Why?
General Chromatin Accessibility	Broad/Narrow	50M	Ideally 3+	Minimum 2	No	- ATAC-seq requires more reads than ChIP since it captures all open-chromatin regions. - Replicates and higher sequencing depth improves statistical power
Active Regulatory Elements	Narrow	50M-100M	Ideally 3+	Minimum 2	No	Enhancers and promoters can be dynamic with weak signals, requiring higher sequencing depth and more replicates.
TF Footprinting	Narrow	100M+	Ideally 3+	Minimum 2	No	TF footprints often have a very small signal size and require single nucleotide resolution, so read depth must be very high.
Differential peaks	Broad or narrow	75M+	Ideally 3+	Minimum 2	No	Differences between conditions can be very subtle, requiring a higher number of reads

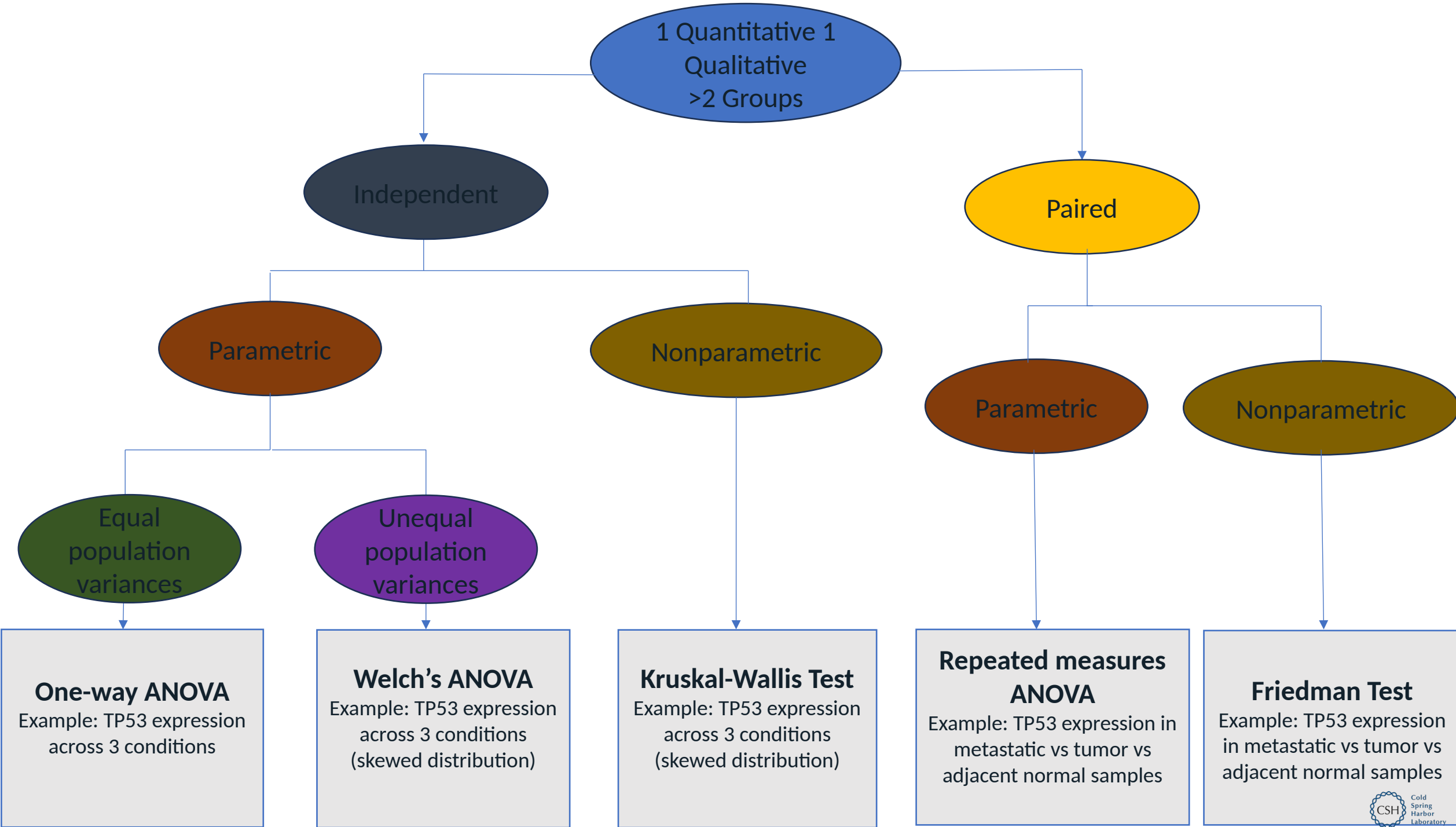












References

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- *Statistics flow chart*
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